

Using PhaserJS to Speed Up 2D Game Development

This tutorial will help you build a 2D platform arcade style game, using the PhaserJS framework, within a few minutes.



PhaserJS is a popular, free and open source HTML5 framework for game development on mobile and desktop browser platforms. It uses HTML5 Canvas for rendering and also WebGL if the device or browser supports it. It is one of the most popular and easy-to-learn game development frameworks and is based on pure Web technologies such as HTML, CSS and JavaScript. With over 15,000 stars and almost 5000 forks at the time of writing this article, it is also one of the most popular and actively maintained game frameworks on GitHub.

Thousands of developers all around the world use Phaser, as it has some great features which are listed below.

Supports JavaScript and TypeScript: PhaserJS supports both JavaScript and TypeScript; so either of these languages can be used to develop games.

Uses WebGL and Canvas: PhaserJS uses both WebGL and Canvas internally, and has the ability to switch between them based on browser support. This allows fast and robust rendering across various mobile and desktop platforms.

Optimised for mobile browsers: PhaserJS was built keeping the mobile Web browser as the target platform, so it is highly optimised for mobile as well as for desktop browsers.

Built-in physics engines: PhaserJS has three built-in physics engines — Arcade Physics, a lightweight engine useful for developing arcade style games; Ninja Physics for advanced tile support; and p2.js, which supports character rigging, constraints and advanced polygon support.

Assets preloader: Assets, images, sprites, sheets, sounds, tilemaps, JSON data, XML, etc., are loaded with ease, parsed and handled automatically—to be used in games and stored in the global cache.

Particles, animation and sprites: PhaserJS has a built-in particle system, which allows developers to easily create cool particle effects such as rain, fire, snow and water. It supports Flash CS6/CC JSON files to create animation for sprites.

Input support: It supports multi-touch, mouse and keyboard. It also allows the user to code custom gesture control.

Plugin system: PhaserJS allows developers to create their own plugins for their games and share them with the community.

Games created using PhaserJS depend on and require a modern browser that supports the Canvas tag, which includes Chrome, Internet Explorer 9+, Firefox, Safari and Opera. It also works on mobile Web browsers such as stock Android 2.x browsers and later, and Mobile Safari for iOS5 and later versions.

Tip: The PhaserJS framework is open source and the source code of the framework is available on GitHub. Developers/programmers are encouraged to participate and contribute to the project.

Installing the prerequisites

Before we start developing our game using PhaserJS, we need to install Node.js, as the PhaserJS framework is also distributed as an npm package and this makes it easy to set up the project structures to develop games.

Note: It is assumed that you have some basic knowledge of Web technologies like HTML, CSS and JavaScript. If you don't, W3Schools (<http://www.w3schools.com/>) is a good place to start. The site has some great tutorials for Web technologies that are easy to follow.